Rahul Rustagi

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Summary

I am a master's student with experience in robotics, computer vision, and autonomous systems, currently immersed in conducting downstream research at Georgia Tech and carrying a strong research background from IIT Kanpur.

Education

MS **Georgia Institute of Technology**, Electrical and Computer Engineering — Aug 2024 – May 2026

- GPA: 4.0/4.0 | *Concentration: Robotics and Perception*
- Coursework: Multi-Robot Systems, Deep Learning for Robotics, PDEs in Image Processing, Computer Vision, Non Linear Systems, Networked and Distributed Control
- Responsibilities: Graduate Teaching Assistant

BS **Indian Institute of Technology (IIT) Kanpur**, Aerospace Engineering — Aug 2020 – May 2024

- GPA: 9.18/10.0 | *Concentration: Aerial Robotics and Machine Learning*
- Coursework: Probabilistic Machine Learning, Reinforcement Learning, Optimal Control, Dynamics, Embedded Cyber-Physical Systems, DSA
- Awards: Academic Excellence (Deans List)
- Responsibilities: Team Head at Aerial Robotics IIT Kanpur

Research Experience

Robot Autonomy Interactive Learning Lab – Georgia Tech, PI: Dr. Sonia Chernova — Atlanta, GA
Project Title: 3D Scene Understanding & Object Tracking — Jan 2025 – Present

- Collaborated with **Amazon Lab126** to design a human-like perception algorithm enabling robots to interpret dynamic environments using topological **Scene Graphs**
- Implemented **Khronos** on **Stretch RE2 Robot** to develop a spatio-temporal scene graph combined with **ORBSLAM2** and **ekf2** for stable odometry
- Working on a **Long-Term Memory Retention Model** for mimicking human memory model for Object Search tasks and performing object association using **Continuous Scene Representations**

Intelligent Vision and Automation Lab– Georgia Tech, PI: Dr. Patricio A. Vela — Atlanta, GA
Project Title: Robot-Agnostic Advanced Navigation Assistance System (ADAS) — Aug 2024 – Jan 2025

- Architected a **hierarchical sensor fusion pipeline** (IMU, LiDAR, RGB-Vision) for real-time visual odometry, achieving **<0.1m APE error** in dynamic environments
- Implemented **GTSAM factor graph** for sensor fusion, improving pose robustness by **40%** under sensor dropout scenarios evaluated **Hessian** matrix to analyze vision pose controllability in X, Y directions
- Integrated **ORBSLAM2** for slow high-precision loop closure with **DSOL (Direct Sparse Odometry and Localization)** for rapid local navigation, enabling navigation in unstructured terrains
- Validated system performance in gazebo simulation on **RotorS** quadrotor and on hardware with **TurtleBot3**, demonstrating robot-agnostic adaptability

Helicopter and VTOL Laboratory – IIT Kanpur, PI: Dr. Abhishek — Kanpur, India
Project Title: Vision-Based Autonomous Quadrotor Landing on Moving Ship — Aug 2023 – May 2024

- Published a **hardware-tested** pipeline to predict landing platform 6DoF motion and achieving touchdown, leveraging deep learning for UAV trajectory planning
- Used **Fractal ArUco markers** for estimating platform 6DoF pose at 60Hz using RGB stream from Intel **Realsense D435i** camera
- Deployed a **LSTM model** on **Jetson Nano TX2**, integrating **TF-TRT & quantization** to predict future pose of platform and iteratively predicted future **2 seconds** of platform 6DoF pose at **20Hz** allowing to plan an optimal landing window time
- Applied **QP Solver** calculating future 2s of optimal time-constrained trajectory way-points in 0.1s for a real safe touchdown
- Tested the pipeline by landing a custom medium-duty quadrotor with **PX4** autopilot support on a **2PRS-1PRU manipulator** emulating ship-wave motion upto seastate 6

Wireless Sensor Network and IoT Laboratory, PI: Dr. Rajesh Hegde — Kanpur, India
Project Title: Using RL for traveling towards optimal wireless charging node in IoT network — Aug 2022 – May 2023

- Designed a **reinforcement learning (RL) framework** to optimize priority-based charging schedules in **low-power IoT networks**, addressing energy constraints
- Engineered a **custom vectorized Gym environment** using **PyBullet physics** to simulate a decentralized network of **10 IoT nodes**, enabling parallel training and scalable policy evaluation
- Benchmarked **TD3-PG, DDPG, and PPO algorithms**, with **TD3-PG** outperforming others by converging **20% faster** to optimal policies and achieving **35% higher cumulative rewards** in sparse-reward environments
- Published methodology in *IEEE TCAS-II* and *IEEE WF-IoT*, demonstrating applicability to smart city sensor networks and industrial IoT deployments

Publications

Vision-Based Autonomous Ship Deck Landing of Unmanned Aerial Vehicle using Fractal ArUco Marker — 2025
C Prachand, ***Rahul Rustagi***, R Shankar, J Singh, A Abhishek, K.S. Venkatesh
10.2514/6.2025-2345 | AIAA SciTech Forum: Unmanned Aerial Systems Track

Lifetime Improvement in Rechargeable Mobile IoT Networks Using Deep Reinforcement Learning — 2024
Aditya Singh, ***Rahul Rustagi***, Rajesh M. Hegde
10.1109/TCSII.2024.3370686 | IEEE Transactions on Circuits and Systems II: Express Briefs

Mobile Energy Transmitter Scheduling in Energy Harvesting IoT Networks using Deep Reinforcement Learning — 2023
Aditya Singh, ***Rahul Rustagi***, Surender Redhu, Rajesh M. Hegde
10.1109/WF-IoT54382.2022.10152078 | IEEE 8th World Forum on Internet of Things

Robotics Projects

Multi-Scale Image Restoration & Motion Estimation via Nonlinear Diffusion — 2025 – Present

- Developed a **Perona-Malik diffusion** model with Laplacian pyramid decomposition to enable noise reduction while preserving critical edge details.
- Leveraged geometric heat equations to reduce edge blurring by 20%.
- Formulated a variational energy functional for **optical flow** (Horn-Schunck) improving accuracy by 12%.

Deep Learning in Computer Vision — CS6476 / GaTech, Aug 2024 – Dec 2024

- Designed a Vision-Language Model using **CLIP** architecture trained on 20% of CIFAR-10 dataset using contrastive learning and aggressive augmentation.
- Implemented **SfM** using SIFT features & epipolar geometry for 6DoF pose estimation.
- Reconstructed 3D scene geometry with less than 5% reprojection error using **COLMAP**.

Perception-Based Intelligent Robot Localization — Carleton University, May 2023 – Dec 2023

- Boosted **JetAuto** robot localization reliability in dynamic environments by deploying **Bayesian sensor fusion** (Kalman filter variant), reducing positional drift by **20%** under sensor noise.
- Refined **Adaptive Monte Carlo Localization (AMCL)** estimates by integrating **g2o** graph optimization with real-time vision feedback

Vision-Based Robotic Hand Dexterity using Inverse Kinematics (IK) — IIT Kanpur / Robotics Club, Mar 2023 – Jul 2023

- Developed an autonomous pipeline for robotic hand object pickup using RGB-D stereo vision and Inverse Kinematics.
- Implemented SE(3) transformations with **tf2_ros** and utilized **MoveIt** for end-effector planning.

Vision-Guided Payload Pickup-Delivery using Drone — IIT Madras / Flipkart GRID 4.0, Nov 2022 – Jan 2023

- Developed an autonomous ROS pipeline for UAV payload pickup and drop-off.
- Employed grid-search with **QGroundControl** and **OpenCV** for 6DoF pose estimation using ArUco markers.

Positions of Responsibility

Graduate Teaching Assistant – CS3630 (Introduction to Perception and Robotics) — Jan 2025 – May 2025

- Handling a total class of 600 students along with 20 other GTA students. Managing **Piazza** discussion forum and responsible for creating projects and making quizzes for students
- Using **WeBots** simulator to create projects on visual navigation and teaching concepts on **bayesian modelling**

Team Head: Software | Aerial Robotics – IIT Kanpur — May 2022 – May 2023

- **Awarded Bronze Medal** in the Drona Pluto Swarm Challenge and secured a spot in the Final Round of Robotics Flipkart Grid 4.0, among top national teams.
- Led and mentored a **cross-functional team of 5 members** to design, develop, and maintain the **software stack** for a fleet of custom-built autonomous aerial robots.
- Spearheaded **full-cycle software development**, including real-time control systems, path planning algorithms, and swarm coordination logic for competition-ready UAVs.

Technical Skills

Robotics Middleware: ROS1/ROS2, Webots, Gazebo, OpenCV, RealSense SDK, PX4, ArduPilot, MAVROS, SITL, HITL

Algorithms & Optimization: VLA Models, YOLO, SfM, ViO, SIFT, ORBSLAM2/3, Segment Anything Model, mIoUs

Programming / Frameworks: Python, C++, TensorFlow, PyTorch, MATLAB, IsaacSim, PyBullet, Eigen, SolidWorks, MoveIt